Modelling intensification in New Zealand English data

Dr. Martin Schweinberger
www.martinschweinberger.de

Habilitation project
Acquisition, Variation, and Diachronic Development of Intensification in English

- synchronic quantitative corpus–based study
- adjectival intensification in New Zealand English (NZE)
- based on the New Zealand component of the *International Corpus of English* (ICE-NZ)

Intensification

(1) yeah... just it would make it so awkward eh you know (ICE-NZ S1A-001:1$M)

(2) um... sara's got a really nice sleeveless green... you know coat jacket (ICE-NZ S1A-002:1$Q)

(3) she was a very nervous sort of a woman (ICE-NZ S1A-018:1$A)

Intensification

Intensification is related to the semantic category of *degree* (degree adverbs) and ranges between very low intensity (downtoning) and very high (amplifiers) (**?**: 589–590).

- Amplifiers (**?**)
  - Maximizers (e.g. *completely*)
  - Boosters (e.g. *very much*)
- Downloners
  - Approximators (e.g. *almost*)
  - Compromisers (e.g. *more or less*)
  - Diminishers (e.g. *partly*)
  - Minimizers (e.g. *hardly*)

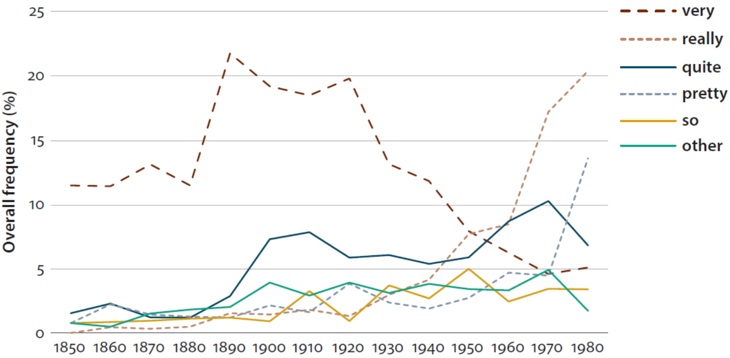
Previous Research

- Intensification...
  - major area of grammatical change (cf. **?**: 441)
  - crucial for the "social and emotional expression of speakers" (**?**: 258)
  - teenage talk and young(ish) speakers (**??**)
  - female speakers (**???**)

Previous Research

- Ongoing changes are accompanied by ...
  - gender and age differences (apparent time construct)
  - differences in the syntactic function (predicative vs attributive)
  - the semantic type of the modified adjective
  - emotional value of the modified adjective (emotional vs non–emotional)
- Intensifying *really* replaces *very* (lexical replacement) (cf. **????**)

Previous study of intensification in New Zealand English (?)



(?: 468)

Research Question

Q_1

Is the NZE Intensifier system currently undergoing change?

ICE New Zealand

ICE New Zealand

New Zealand component of the *International Corpus of English* (?)

- released in 1999 (*The Victoria University of Wellington*)
- consists of one million words (600,000 spoken and 400,000 written)
- representing diverse spoken and written text types
- here only private dialogues (200,000 words)

Data Processing

Data Processing

- Split spoken data into utterances
- Removal of meta information
- Part–of-speech tagging
- Retrieving adjectives (PoS–tag JJ)
- Determining whether adjective is preceded by an intensifying adverb (PoS–tag RB)

Data Processing

- Determining the syntactic type of adjective (predicative vs attributive (if followed by NN* tag))
- Removal of
  - negated adjectives
  - comparative and superlative forms
  - non–intensifiable forms (categorical, e.g. nationalities | locations: *asian*, *Asia*)
- Sentiment Analysis determines the emotional value of adjectives based on the *Word-Emotion Association Lexicon* (?)
- Manual cross–evaluation of automated classification
- Adding speaker information (age, sex, etc.).

Data Summary

Data Summary: ICE-NZ data

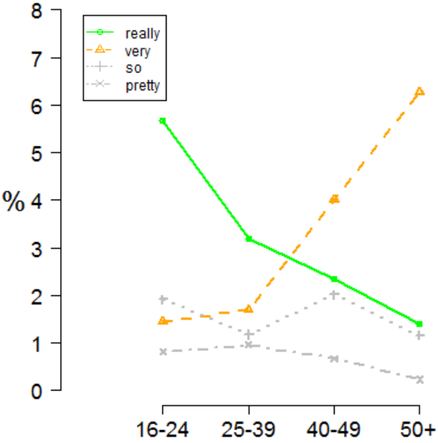
Age	**Sex**	**Speakers** (N)	**Adj.** (N)	**Int.** (N)	**Int.** (%)
16-24	female	39	1102	140	12.7
16-24	male	29	811	81	10.0
25-39	female	23	629	65	10.3
25-39	male	16	481	35	7.3
40-49	female	16	509	60	11.8
40-49	male	9	172	7	4.1
50+	female	7	259	27	10.4
50+	male	6	236	25	10.6
Total		145	4199	440	10.5

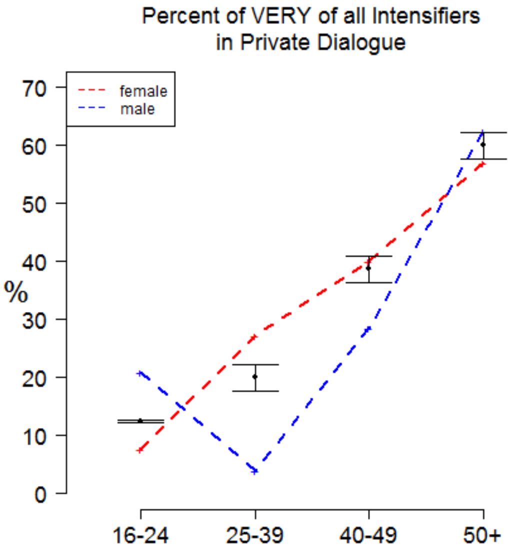
Data Summary: Intensifiers ICE-NZ

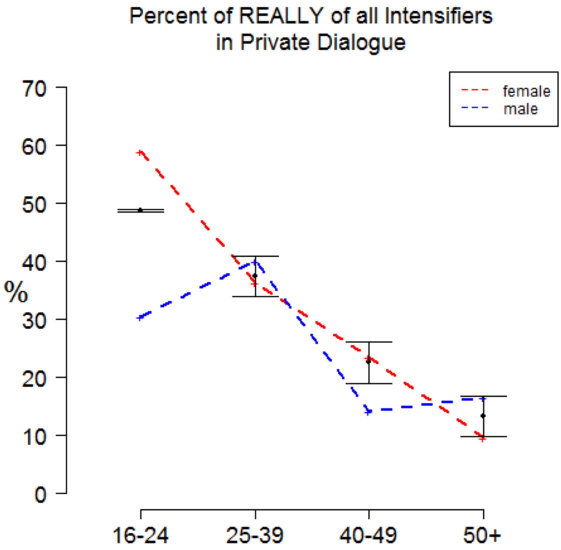
Intensifier	**N**	**%**	**Int.** (%)
∅ Intensification	3759	89.52	
really	150	3.57	34.09
very	96	2.29	21.82
so	66	1.57	15.00
too	34	0.81	7.73
pretty	29	0.69	6.59
real	18	0.43	4.09
well	7	0.17	1.59
absolutely, right, totally	5	0.36	3.42
bloody	4	0.10	0.91
crazy, particularly	2	0.10	0.90
actually, badly, completely, definitely, dreadfully, enormously, entirely, excruciatingly, fucking, fully, horrendously, incredibly, obviously, purely, shocking, true, wicked	1	0.34	3.91
Total	4199	10.48	100

Results
—
Visualization

Intensifiers across Age Cohorts




Percent of VERY of all Intensifiers
in Private Dialogue
female
male
%
70
60
50
40
30
20
10
0
16-24
25-39
40-49
50+


Percent of REALLY of all Intensifiers
in Private Dialogue
female
male
70
60
50
40
%
30
20
10
0
16-24
25-39
40-49
50+

Statistical Analysis

Research question

Q_2

Which factors correlate with the use of *really* (innovation)?
(age, sex, syntactic function, . . .)

Statistical Analysis

- Mixed–effects binomial logistic regression models
  - AIC based, step-wise step-up model fitting

Dependent Variable(s)

really	nominal	yes/no occurrence of pre-adjectival *really*

Independent Variable(s)

age	categorical	age groups in ascending order	extra	linguistic
sex	nominal	male | female		
eth	nominal	pakeha | maori		
occ	nominal	acmp | sml		
emo	nominal	emotional | nonemotional	intra	linguistic
fun	nominal	attributive | predicative		
sem	categorical	semantic type of adjective		
grad	nominal	gradable | nongradable		

Regression Results

Regression Results

	Group(s)	Variance	Std. Dev.	L.R.χ^2(df1)	Sig.
Random Effect(s)	flid	0.44	0.66	29	p<.001***
Fixed Effect(s)	**Estimate**	**VIF**	**OddsRatio**	**z value**	**Sig.**
(Intercept)	-5.04		0.01	-14.55	p<.001***
age:25-39	-0.57	1.07	0.57	-2.09	p<.05*
age:40-49	-0.94	1.08	0.39	-2.7	p<.01**
age:50+	-1.48	1.03	0.23	-2.98	p<.01**
sex:male	-0.85	1.01	0.43	-3.46	p<.001***
fun:predicative	0.74	1	2.09	4.09	p<.001***
grad:nograd	1.88	1.01	6.52	6.31	p<.001***
emo:emotional	0.79	1.01	2.21	4.49	p<.001***
Model statistics					**Value**
Number of Groups					145
Cases in model					4199
Observed successes					150
R^2 (Nagelkerke)					0.155
C					0.844
Somers' D_{xy}					0.688
Prediction accuracy					96.43%
Model LL Ratio Test			L.R.χ^2(df8)	176.67	p<.001***

Regression Results

	Group(s)	Variance	Std. Dev.	L.R.χ^2(df1)	Sig.
Random Effect(s)	flid	0.44	0.66	29	p<.001***
Fixed Effect(s)	**Estimate**	**VIF**	**OddsRatio**	**z value**	**Sig.**
(Intercept)	-5.04		0.01	-14.55	p<.001***
age:25-39	-0.57	1.07	0.57	-2.09	p<.05*
age:40-49	-0.94	1.08	0.39	-2.7	p<.01**
age:50+	-1.48	1.03	0.23	-2.98	p<.01**
sex:male	-0.85	1.01	0.43	-3.46	p<.001***
fun:predicative	0.74	1	2.09	4.09	p<.001***
grad:nograd	1.88	1.01	6.52	6.31	p<.001***
emo:emotional	0.79	1.01	2.21	4.49	p<.001***
Model statistics					**Value**
Number of Groups					145
Cases in model					4199
Observed successes					150
R^2 (Nagelkerke)					0.155
C					**0.844**
Somers' D_{xy}					**0.688**
Prediction accuracy					**96.43%**
Model LL Ratio Test			L.R.χ^2(df8)	176.67	p<.001***

Regression Results

	Group(s)	Variance	Std. Dev.	L.R.χ^2(df1)	Sig.
Random Effect(s)	flid	0.44	0.66	29	p<.001***
Fixed Effect(s)	**Estimate**	**VIF**	**OddsRatio**	**z value**	**Sig.**
(Intercept)	-5.04		0.01	-14.55	p<.001***
age:25-39	-0.57	1.07	0.57	-2.09	p<.05*
age:40-49	-0.94	1.08	0.39	-2.7	p<.01**
age:50+	-1.48	1.03	0.23	-2.98	p<.01**
sex:male	-0.85	1.01	0.43	-3.46	p<.001***
fun:predicative	0.74	1	2.09	4.09	p<.001***
grad:nograd	1.88	1.01	6.52	6.31	p<.001***
emo:emotional	0.79	1.01	2.21	4.49	p<.001***
Model statistics					**Value**
Number of Groups					**145**
Cases in model					**4199**
Observed successes					**150**
R^2 (Nagelkerke)					**0.155**
C					0.844
Somers' D_{xy}					0.688
Prediction accuracy					96.43%
Model LL Ratio Test			L.R.χ^2(df8)	176.67	p<.001***

Regression Results

	Group(s)	Variance	Std. Dev.	L.R.χ^2(df1)	Sig.
Random Effect(s)	flid	0.44	0.66	29	p<.001***
Fixed Effect(s)	**Estimate**	**VIF**	**OddsRatio**	**z value**	**Sig.**
(Intercept)	-5.04		0.01	-14.55	p<.001***
age:25-39	-0.57	1.07	**0.57**	-2.09	p<.05*
age:40-49	-0.94	1.08	**0.39**	-2.7	p<.01**
age:50+	-1.48	1.03	**0.23**	-2.98	p<.01**
sex:male	-0.85	1.01	0.43	-3.46	p<.001***
fun:predicative	0.74	1	2.09	4.09	p<.001***
grad:nograd	1.88	1.01	6.52	6.31	p<.001***
emo:emotional	0.79	1.01	2.21	4.49	p<.001***
Model statistics					**Value**
Number of Groups					145
Cases in model					4199
Observed successes					150
R^2 (Nagelkerke)					0.155
C					0.844
Somers' D_{xy}					0.688
Prediction accuracy					96.43%
Model LL Ratio Test			L.R.χ^2(df8)	176.67	p<.001***

Regression Results

	Group(s)	Variance	Std. Dev.	L.R.χ^2(df1)	Sig.
Random Effect(s)	flid	0.44	0.66	29	p<.001***
Fixed Effect(s)	**Estimate**	**VIF**	**OddsRatio**	**z value**	**Sig.**
(Intercept)	-5.04		0.01	-14.55	p<.001***
age:25-39	-0.57	1.07	0.57	-2.09	p<.05*
age:40-49	-0.94	1.08	0.39	-2.7	p<.01**
age:50+	-1.48	1.03	0.23	-2.98	p<.01**
sex:male	-0.85	1.01	**0.43**	-3.46	p<.001***
fun:predicative	0.74	1	2.09	4.09	p<.001***
grad:nograd	1.88	1.01	6.52	6.31	p<.001***
emo:emotional	0.79	1.01	2.21	4.49	p<.001***
Model statistics					**Value**
Number of Groups					145
Cases in model					4199
Observed successes					150
R^2 (Nagelkerke)					0.155
C					0.844
Somers' D_{xy}					0.688
Prediction accuracy					96.43%
Model LL Ratio Test			L.R.χ^2(df8)	176.67	p<.001***

Regression Results

	Group(s)	Variance	Std. Dev.	L.R.χ^2(df1)	Sig.
Random Effect(s)	flid	0.44	0.66	29	p<.001***
Fixed Effect(s)	**Estimate**	**VIF**	**OddsRatio**	**z value**	**Sig.**
(Intercept)	-5.04		0.01	-14.55	p<.001***
age:25-39	-0.57	1.07	0.57	-2.09	p<.05*
age:40-49	-0.94	1.08	0.39	-2.7	p<.01**
age:50+	-1.48	1.03	0.23	-2.98	p<.01**
sex:male	-0.85	1.01	0.43	-3.46	p<.001***
fun:predicative	0.74	1	**2.09**	4.09	p<.001***
grad:nograd	1.88	1.01	**6.52**	6.31	p<.001***
emo:emotional	0.79	1.01	**2.21**	4.49	p<.001***
Model statistics					**Value**
Number of Groups					145
Cases in model					4199
Observed successes					150
R^2 (Nagelkerke)					0.155
C					0.844
Somers' D_{xy}					0.688
Prediction accuracy					96.43%
Model LL Ratio Test			L.R.χ^2(df8)	176.67	p<.001***

Summary

- The NZE intensifier system is undergoing change
- The observed change is accompanied by heavy stratification
- Semantic and pragmatic factors are the best predictors for the use of *really*

But why is *really* taking over???

Collocates

Q_3

Why and how is *really* replacing *very*?
Why not another variant?

H_1

Successful variants (*really*) associate with many and particularly high frequency adjectives.

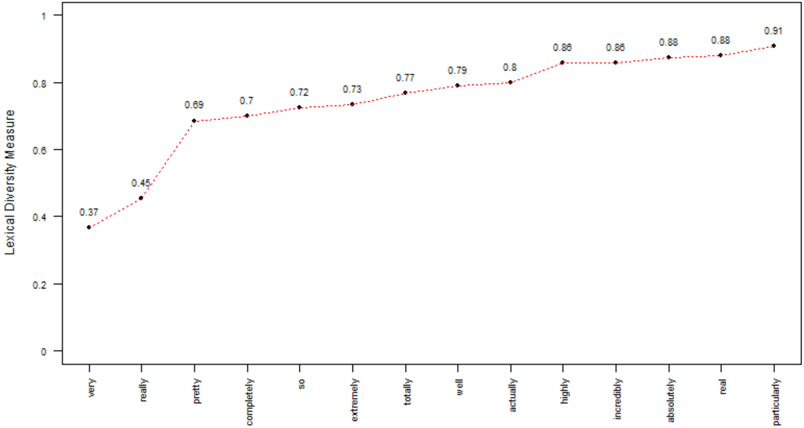


Figure: adjective types/intensifier tokens (ICE NZ)

Adjective	Intensifier	Age 16-24	25-39	40-49	50+
difficult	really	0	2	0	0
good	really	27	9	2	3
hard	really	5	3	0	1
important	really	0	1	1	2
interesting	really	1	2	1	2
little	really	0	0	0	0
nice	really	7	1	1	1
strong	really	0	0	0	0
difficult	very	0	4	5	12
good	very	5	15	20	23
hard	very	0	3	4	4
important	very	2	6	4	8
interesting	very	1	0	2	8
little	very	0	5	4	11
nice	very	3	2	5	3
strong	very	0	2	4	8
difficult	other	2	3	3	2
good	other	8	6	6	6
hard	other	5	2	2	0
important	other	1	0	2	2
interesting	other	1	1	0	1
little	other	0	0	1	1
nice	other	0	0	0	0
strong	other	0	3	0	0

increasing trend
receding trend

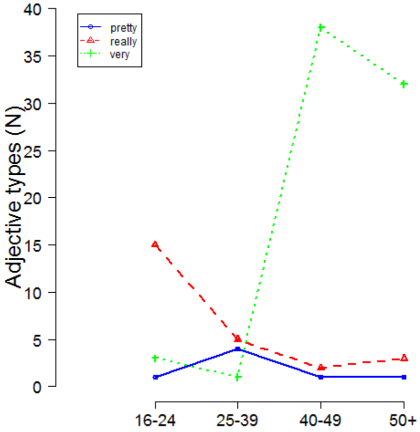


Figure: intensifier types : adjective types : age (ICE NZ)

Covarying Collexeme Analysis (CCA)

- Member of the *collostructional analysis* family that aims at measuring the degree of attraction of lemmas in one slot of a construction to lemmas in another slot of the same construction. (?)
- CCA differs from most collocation statistics such that it takes syntactic structure into account and it applies Fisher-Yates exact test and is thus not based on any distributional assumptions.

Collocations by Age

Age	**Adjective**	**Intensifier**	**OddsRatio**	**Bonf. Corr. Sig**
16-24	really	good	5.44	p<.001
50+	very	difficult	20.07	p<.001
50+	very	good	4.72	p<.001
50+	very	strong	21.33	p<.01

Correspondence Analysis (CA)

- Exploratory method for correspondence between rows and column values/counts in distributional data (similar to factor analysis) (**?**: 128–136)
- Restrictions on intensifiers
  Minimum of 7 intensifier tokens and intensifier type has to collocate with more than 1 adjective type
- Restrictions on adjectives
  Minimum of 7 adjective tokens

	pretty	real	really	so	very
bad	1	2	3	1	1
clear	0	1	0	0	7
close	0	0	2	2	3
different	0	0	1	2	4
difficult	2	0	2	2	21
first	0	0	0	0	9

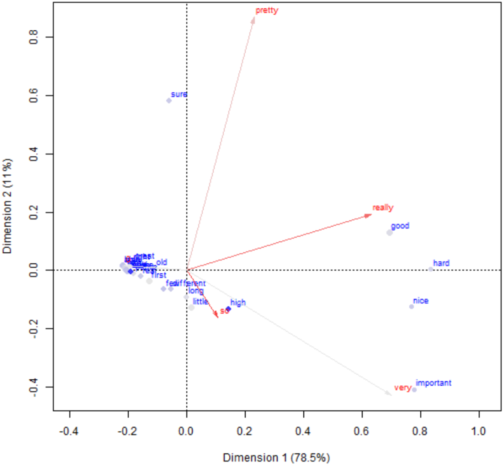


Figure: Correspondence of intensifier and adjective tokens (NZE)

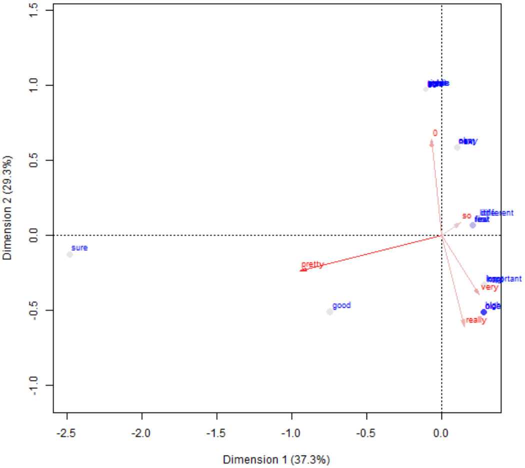


Figure: Correspondence of intensifier and adjective types (NZE)

Semantic Vector Space Models (SVSM)

- Distributional approach to semantics (cf. ?: 323–331).
- Words that share collocates are semantically similar (?: 368–370).
- Semantic similarity can thus be measured in collocate frequency.
- Collocate frequency is captured by cross-tabulating frequency counts.

	extremely	pretty	real	really	so	totally	very
basic	0	1	1	0	0	0	1
black	0	0	0	0	0	1	1
brave	0	0	0	0	0	0	1
busy	0	1	0	1	1	0	1
careful	0	0	0	1	0	0	1
clear	0	0	1	0	0	0	1

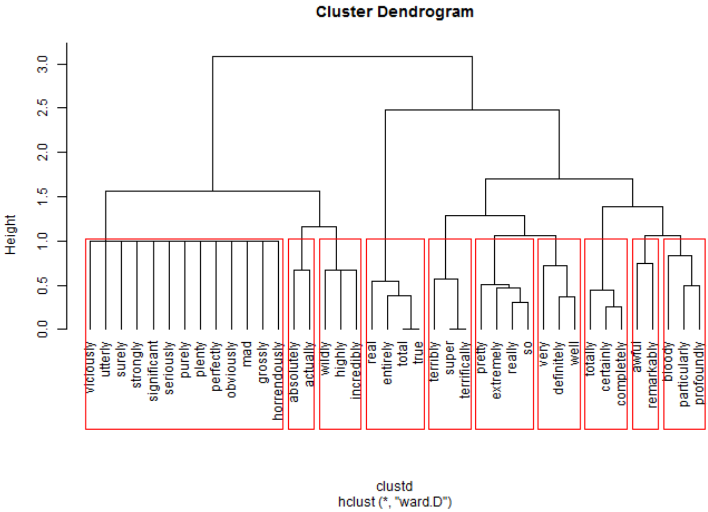
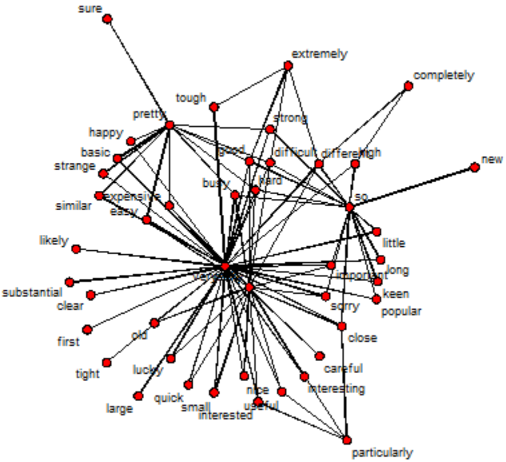


Figure: Clustering of intensifiers by adjective collocates (NZE)


sure
extremely
completely
tough
pretty
strong
happy
basic
good
difficult
high
new
strange
busy
hard
similar
expensive
so
easy
likely
little
long
important
substantial
keen
clear
sorry
popular
first
old
close
tight
lucky
careful
quick
interesting
large
small
nice
useful
interested
particularly

SUMMARY

Intensifying *really*

- use declines almost linearly with age (incoming innovation)
- is dis-preferred by male speakers (female dominated change)
- collocates with adjectives that are emotional
- used preferentially in predicative function
- is preferred by non-gradual adjectives

Really is heavily stratified and correlates with extra-linguistic factors (age, sex) and even more strongly with intra-linguistic factors (emotionality, gradability, syntactic function).

Successful intensifiers (*really*)

- increase the number of adjectives that they co-occur with
- collocate significantly with high frequency adjectives (*good* among 16-24 year olds)
- semantically similar to dominant variant (have similar collocation profiles)

Successful intensifier variants collocate with many adjectives, particularly high frequency adjectives, and show collocation profiles similar to those of dominant variants.

Problems & Outlook

Problems

- apply multinominal regression instead of binomial (dependent variable is categorical, really, very, other, null)
- use Wellington corpus instead of ICE (1,000,000 instead of 200,000 words) → increase of data base
- apply SVSM to adjectives to find distinct types of adjectives
- define variable context and variants better (not all adjectives take all intensifiers)!
- clean data: find true variants ($^?$*completely* good : *very* good)

THANK YOU SO, REALLY, VERY MUCH!

Baayen, R. H. (2008). *Analyzing linguistic data. A practical introduction to statistics using R*. Cambridge: Cambridge University press.

Bauer, L. and W. Bauer (2002). Adjective boosters in the english of young new zealanders. *Journal of English Linguistics 30*, 244–257.

Bauer, L., A. Bell, D. Britain, G. Kennedy, C. Lane, M. Meyerhoff, and M. Stubbe (1999). The new zealand component of the international corpus of english (ice-nz).

Brinton, L. J. and L. K. Arnovick (2006). *The English Language: A Linguistic History*. Oxford: Oxford University Press.

D'Arcy, A. F. (2015). Stability, stasis and change – the longue durée of intensification. *Diachronica 32*(04), 449–493.

Ito, R. and S. Tagliamonte (2003). Well weird, right dodgy, very strange, really cool: Layering and recycling in english intensifiers. *Language in Society 32*, 257—279.

Levshina, N. (2015). *How to do linguistics with R: Data exploration and statistical analysis*. Amsterdam: John Benjamins Publishing Company.

Macaulay, R. (2006). Pure grammaticalization: The development of a teenage intensifier. *Language Variation and Change 18*, 267—283.

Mohammad, S. M. and P. D. Turney (2013). Crowdsourcing a word–emotion association lexicon. *Computational Intelligence 29*(3), 436–465.

Quirk, R., S. Greenbaum, G. Leech, and J. Svartvik (1985). *A Comprehensive Grammar of the English Language*. London & New York: Longman.

Stefanowitsch, A. (2010). Empirical cognitive semantics: Some thoughts. In D. Glynn and K. Fischer (Eds.), *Quantitative Cognitive Semantics: Corpus-driven approaches*, Volume 46, pp. 355–380. Walter de Gruyter.

Stefanowitsch, A. and S. T. Gries (2005). Covarying collexemes. *Corpus linguistics and linguistic theory 1*(1), 1–43.

Tagliamonte, S. (2005). So who? like how? just what?: Discourse markers in the conversations of young canadians. *Journal of Pragmatics 37*(11), 1896–1915.

Tagliamonte, S. (2006). "so cool, right?": Canadian english entering the 21st century. *The Canadian Journal of Linguistics/La revue canadienne de linguistique 51*(2), 309–331.

Tagliamonte, S. (2008). So different and pretty cool! recycling intensifiers in toronto, canada. *English Language and Linguistics 12*(2), 361–394.

Modelling intensification in New Zealand English data

Dr. Martin Schweinberger
www.martinschweinberger.de